

HW3

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Q1: 1.3

(i)

Show that $\mathcal{F}_6 = \{A \subset \Omega : A \text{ is countable or } A^c \text{ is countable}\}$ from Example 1.1.4 is a σ -algebra.

Answer

Proof. Let Ω be a nonempty set.

- (a) $\Omega^c = \emptyset$, which is countable (even if Ω is not). Thus, $\Omega \in \mathcal{F}_6$.
- (b) Let $A \in \mathcal{F}_6$ be given.

Case 1: A is countable. Then $(A^c)^c = A$ is countable, so $A^c \in \mathcal{F}_6$.

Case 2: A^c is countable. Then $A^c \in \mathcal{F}_6$ by the definition of $\mathcal{F}_6$.

In either case, $A^c \in \mathcal{F}_6$.

- (d) Let $\{A_n\}_{n \geq 1} \subset \mathcal{F}_6$ be given, and set $A \equiv \bigcup_{n \geq 1} A_n$.

Case 1: A_n is countable for all $n \geq 1$. Then A is a countable union of countable sets, and hence is countable. Thus, $A \in \mathcal{F}_6$.

Case 2: A_{n_0} is uncountable for some $n_0 \in \mathbb{N}$. Since $A_{n_0} \in \mathcal{F}_6$, $A_{n_0}^c$ must be countable. By De Morgan's law,

$$A^c = \left(\bigcup_{n \geq 1} A_n \right)^c = \bigcap_{n \geq 1} A_n^c \subset A_{n_0}^c.$$

Since any subset of a countable set is countable, A^c is countable. Thus, $A \in \mathcal{F}_6$.

- (c) By (a) and (b), $\emptyset = \Omega^c \in \mathcal{F}_6$. Let $A, B \in \mathcal{F}_6$ be given, and take $A_1 = A$, $A_2 = B$, and $A_n = \emptyset$ for all $n \geq 3$. Then by (d), $A \cup B = \bigcup_{n \geq 1} A_n \in \mathcal{F}_6$.

Thus, by (a)–(c), $\mathcal{F}_6$ is an algebra, and by (d), $\mathcal{F}_6$ is a σ -algebra. //

(ii)

Define μ on $\mathcal{F}_6$ as $\mu(A) = 1$ if A is uncountable and $\mu(A) = 0$ if A is countable. Is μ a measure on $\mathcal{F}_6$? Justify.

Answer

Yes, μ is a measure on $\mathcal{F}_6$.

Proof.

- (a) $\mu(A) \in \{0, 1\} \subset [0, \infty]$ for all $A \in \mathcal{F}_6$, so (a) holds.
- (b) $\emptyset$ is countable, so $\mu(\emptyset) = 0$, and (b) holds.

(c) *Claim:* If $A, B \in \mathcal{F}_6$ with $A \cap B = \emptyset$, then A and B are not both uncountable.

Suppose A is uncountable. Since $A \in \mathcal{F}_6$, A^c is countable. Since $A \cap B = \emptyset$, $B \subset A^c$, and so B is countable.

Now let $\{A_n\}_{n \geq 1} \subset \mathcal{F}_6$ be a disjoint collection, and let $A \equiv \bigcup_{n \geq 1} A_n$. Note that $A \in \mathcal{F}_6$ by (i). By the claim, at most one A_n is uncountable.

Case 1: A_n is countable for all $n \geq 1$. Then A is countable, and

$$\mu(A) = 0 = \sum_{n=1}^{\infty} 0 = \sum_{n=1}^{\infty} \mu(A_n).$$

Case 2: Exactly one A_{n_0} is uncountable. Then $A \supset A_{n_0}$ is uncountable, so $\mu(A) = 1$. Also, $\mu(A_n) = 0$ for all $n \neq n_0$, so

$$\sum_{n=1}^{\infty} \mu(A_n) = \mu(A_{n_0}) + \sum_{n \neq n_0} \mu(A_n) = 1 + 0 = 1 = \mu(A).$$

Thus, (c) holds, and $(\Omega, \mathcal{F}_6, \mu)$ is a measure space. //

Remark. If Ω is countable, then every $A \subset \Omega$ is countable, and $\mu \equiv 0$ is the trivial measure. If Ω is uncountable, then $\mu(\Omega) = 1$, so μ is in fact a probability measure.

Q2: 1.8

Let $\Omega \equiv \{1, 2, \dots\} = \mathbb{N}$ and $A_i \equiv \{j : j \in \mathbb{N}, j \geq i\}$, $i \in \mathbb{N}$. Show that $\sigma\langle \mathcal{A} \rangle = \mathcal{P}(\Omega)$ where $\mathcal{A} = \{A_i : i \in \mathbb{N}\}$.

Answer

Proof.

Part 1: WTS $\sigma\langle \mathcal{A} \rangle \subset \mathcal{P}(\Omega)$.

$A_i \subset \Omega$ for all $i \in \mathbb{N}$, so $\mathcal{A} \subset \mathcal{P}(\Omega)$. Also, $\mathcal{P}(\Omega)$ is a σ -algebra (Example 1.1.2). Thus $\mathcal{P}(\Omega) \in \mathcal{I}(\mathcal{A})$, and so

$$\sigma\langle \mathcal{A} \rangle = \bigcap_{\mathcal{F} \in \mathcal{I}(\mathcal{A})} \mathcal{F} \subset \mathcal{P}(\Omega).$$

Part 2: WTS $\sigma\langle \mathcal{A} \rangle \supset \mathcal{P}(\Omega)$.

Step 1: $\{i\} = A_i \cap A_{i+1}^c$ for all $i \in \mathbb{N}$.

($\subset$) Since $i \geq i$, $i \in A_i$. Since $i < i + 1$, $i \notin A_{i+1}$, i.e., $i \in A_{i+1}^c$. Thus $i \in A_i \cap A_{i+1}^c$.

($\supset$) Let $j \in A_i \cap A_{i+1}^c$. Then $j \geq i$ and $j < i + 1$. Since $j \in \mathbb{N}$, $j = i$.

Step 2: $\{i\} \in \sigma\langle \mathcal{A} \rangle$ for all $i \in \mathbb{N}$.

$A_i, A_{i+1} \in \mathcal{A} \subset \sigma\langle \mathcal{A} \rangle$. Since $\sigma\langle \mathcal{A} \rangle$ is a σ -algebra, it is closed under complementation, so $A_{i+1}^c \in \sigma\langle \mathcal{A} \rangle$. It is also closed under finite intersections, since $E \cap F = (E^c \cup F^c)^c$. Thus, by Step 1, $\{i\} = A_i \cap A_{i+1}^c \in \sigma\langle \mathcal{A} \rangle$.

Step 3: $B \in \sigma\langle \mathcal{A} \rangle$ for all $B \in \mathcal{P}(\Omega)$.

Let $B \in \mathcal{P}(\Omega)$ be given.

Case 1: $B = \emptyset$. Then $B = \Omega^c \in \sigma\langle \mathcal{A} \rangle$.

Case 2: $B \neq \emptyset$. Since $B \subset \mathbb{N}$, B is countable, and

$$B = \bigcup_{i \in B} \{i\}$$

is a countable union of sets in $\sigma\langle\mathcal{A}\rangle$ by Step 2. Since $\sigma\langle\mathcal{A}\rangle$ is closed under countable unions, $B \in \sigma\langle\mathcal{A}\rangle$.
Thus $\mathcal{P}(\Omega) \subset \sigma\langle\mathcal{A}\rangle$, and combining Parts 1 and 2, $\sigma\langle\mathcal{A}\rangle = \mathcal{P}(\Omega)$. //

Q3: 1.10

Show that in Example 1.1.6, $\mathcal{O}_j \subset \sigma\langle\mathcal{O}_i\rangle$ for all $1 \leq i, j \leq 4$.

(Okay to do for $k = 2$, i.e. $\mathbb{R}^2$.)

Note: General Strategy; For classes $\mathcal{D}, \mathcal{E} \subset \mathcal{P}(\mathbb{R}^2)$, $\mathcal{D} \subset \sigma\langle\mathcal{E}\rangle$ implies $\sigma\langle\mathcal{D}\rangle \subset \sigma\langle\mathcal{E}\rangle$, since $\sigma\langle\mathcal{E}\rangle$ is a σ -algebra containing $\mathcal{D}$. So it suffices to establish the cycle

$$\mathcal{O}_1 \subset \sigma\langle\mathcal{O}_3\rangle, \quad \mathcal{O}_3 \subset \sigma\langle\mathcal{O}_4\rangle, \quad \mathcal{O}_4 \subset \sigma\langle\mathcal{O}_2\rangle, \quad \mathcal{O}_2 \subset \sigma\langle\mathcal{O}_1\rangle.$$

Each part below proves the link beginning at its $\mathcal{O}_j$.

(i)

$$\mathcal{O}_1 = \{(a_1, b_1) \times (a_2, b_2) : -\infty \leq a_1 < b_1 \leq \infty, -\infty \leq a_2 < b_2 \leq \infty\}$$

Answer

Proof. Let $R = (a_1, b_1) \times (a_2, b_2) \in \mathcal{O}_1$, and let $J \equiv \{(p_1, q_1, p_2, q_2) \in \mathbb{Q}^4 : a_i < p_i < q_i < b_i, i = 1, 2\}$. We claim

$$R = \bigcup_{(p_1, q_1, p_2, q_2) \in J} (p_1, q_1) \times (p_2, q_2).$$

($\supset$) Each rectangle in the union is contained in R .

($\subset$) Let $(x_1, x_2) \in R$. By the density of $\mathbb{Q}$, choose $p_i \in (a_i, x_i) \cap \mathbb{Q}$ and $q_i \in (x_i, b_i) \cap \mathbb{Q}$. Then $(p_1, q_1, p_2, q_2) \in J$ and $(x_1, x_2) \in (p_1, q_1) \times (p_2, q_2)$.

Since $J \subset \mathbb{Q}^4$ is countable and each $(p_1, q_1) \times (p_2, q_2) \in \mathcal{O}_3$, $R \in \sigma\langle\mathcal{O}_3\rangle$. Thus $\mathcal{O}_1 \subset \sigma\langle\mathcal{O}_3\rangle$. //

(ii)

$$\mathcal{O}_2 = \{(-\infty, x_1) \times (-\infty, x_2) : -\infty < x_1, x_2 < \infty\}$$

Answer

Proof. Let $(-\infty, x_1) \times (-\infty, x_2) \in \mathcal{O}_2$. Taking $a_i = -\infty$ and $b_i = x_i$ in the definition of $\mathcal{O}_1$ (valid since $-\infty < x_i < \infty$), this set is in $\mathcal{O}_1$. Thus $\mathcal{O}_2 \subset \mathcal{O}_1 \subset \sigma\langle\mathcal{O}_1\rangle$. //

(iii)

$$\mathcal{O}_3 = \{(a_1, b_1) \times (a_2, b_2) : a_1 < b_1, a_2 < b_2, a_1, a_2, b_1, b_2 \in \mathbb{Q}\}$$

Answer

Proof. Write $H(x_1, x_2) \equiv (-\infty, x_1) \times (-\infty, x_2)$, so $\mathcal{O}_4 = \{H(x_1, x_2) : x_1, x_2 \in \mathbb{Q}\}$. Let $R = (a_1, b_1) \times (a_2, b_2) \in \mathcal{O}_3$, and let $c_{i,n} \equiv a_i + n^{-1} \in \mathbb{Q}$ for $n \in \mathbb{N}$.

Step 1. For $c_1, c_2 \in \mathbb{Q}$,

$$[c_1, b_1) \times [c_2, b_2) = H(b_1, b_2) \setminus [H(c_1, b_2) \cup H(b_1, c_2)].$$

For $y \in H(b_1, b_2)$, we have $y \notin H(c_1, b_2)$ iff $y_1 \geq c_1$, and $y \notin H(b_1, c_2)$ iff $y_2 \geq c_2$. So both sides equal $\{y : c_i \leq y_i < b_i, i = 1, 2\}$.

All three quadrants are in $\mathcal{O}_4$. Since $\sigma\langle\mathcal{O}_4\rangle$ is closed under finite unions and differences ($E \setminus F = E \cap F^c$), $[c_1, b_1) \times [c_2, b_2) \in \sigma\langle\mathcal{O}_4\rangle$.

Step 2. We claim $R = \bigcup_{n \geq 1} [c_{1,n}, b_1) \times [c_{2,n}, b_2)$.

($\supset$) Since $c_{i,n} > a_i$, each set in the union is contained in R .

($\subset$) Let $y \in R$, and choose n with $n^{-1} < \min_i (y_i - a_i)$. Then $c_{i,n} < y_i < b_i$ for $i = 1, 2$.

By Step 1, R is a countable union of sets in $\sigma\langle\mathcal{O}_4\rangle$, so $R \in \sigma\langle\mathcal{O}_4\rangle$. Thus $\mathcal{O}_3 \subset \sigma\langle\mathcal{O}_4\rangle$. //

(iv)

$$\mathcal{O}_4 = \{(-\infty, x_1) \times (-\infty, x_2) : x_1, x_2 \in \mathbb{Q}\}$$

Answer

Proof. Since $\mathbb{Q} \subset \mathbb{R}$, $\mathcal{O}_4 \subset \mathcal{O}_2 \subset \sigma\langle\mathcal{O}_2\rangle$.

Combining (i)–(iv) with the strategy above,

$$\sigma\langle\mathcal{O}_1\rangle \subset \sigma\langle\mathcal{O}_3\rangle \subset \sigma\langle\mathcal{O}_4\rangle \subset \sigma\langle\mathcal{O}_2\rangle \subset \sigma\langle\mathcal{O}_1\rangle.$$

So all four σ -algebras are equal, and $\mathcal{O}_j \subset \sigma\langle\mathcal{O}_j\rangle = \sigma\langle\mathcal{O}_i\rangle$ for all $1 \leq i, j \leq 4$. //

Q4: 1.18

Let $\Omega \equiv \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and $A_n = \{j : j \in \mathbb{N}, j \geq n\}$, $n \in \mathbb{N}$. Let μ be the counting measure on $(\Omega, \mathcal{F})$. Verify that $\lim_{n \rightarrow \infty} \mu(A_n) \neq \mu(\bigcap_{n \geq 1} A_n)$.

Answer

Proof.

Step 1: $\lim_{n \rightarrow \infty} \mu(A_n) = \infty$.

Let $n \in \mathbb{N}$ be given. The map $j \mapsto n + j - 1$ is an injection from $\mathbb{N}$ into A_n , so A_n is infinite and $\mu(A_n) = |A_n| = \infty$. Since this holds for all n , $\lim_{n \rightarrow \infty} \mu(A_n) = \infty$.

Step 2: $\bigcap_{n \geq 1} A_n = \emptyset$.

($\supset$) Trivial.

($\subset$) Let $j \in \mathbb{N}$ be given. Since $j < j + 1$, $j \notin A_{j+1}$, and hence $j \notin \bigcap_{n \geq 1} A_n$. Thus no element of Ω lies in $\bigcap_{n \geq 1} A_n$.

Step 3. By Step 2 and property (b) of Definition 1.2.1, $\mu\left(\bigcap_{n \geq 1} A_n\right) = \mu(\emptyset) = 0$. Combining with Step 1,

$$\lim_{n \rightarrow \infty} \mu(A_n) = \infty \neq 0 = \mu\left(\bigcap_{n \geq 1} A_n\right). \quad //$$

Q5: 1.19

Let Ω be a nonempty set and let $\mathcal{C} \subset \mathcal{P}(\Omega)$ be a semialgebra. Let

$$\mathcal{A}(\mathcal{C}) \equiv \left\{ A : A = \bigcup_{i=1}^k B_i : B_i \in \mathcal{C}, i = 1, 2, \dots, k, k \in \mathbb{N} \right\}$$

(i)

Show that $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$.

Answer

Proof. Assume $\mathcal{C} \neq \emptyset$. (If $\mathcal{C} = \emptyset$, then $\mathcal{A}(\mathcal{C}) = \emptyset$, and the claim fails trivially.)

Part 1: WTS $\mathcal{C} \subset \mathcal{A}(\mathcal{C})$.

For $B \in \mathcal{C}$, take $k = 1$ and $B_1 = B$. Then $B \in \mathcal{A}(\mathcal{C})$.

Part 2: WTS $\mathcal{A}(\mathcal{C})$ is an algebra.

(a) Pick any $A \in \mathcal{C}$. By the definition of a semialgebra, there exist $B_1, \dots, B_m \in \mathcal{C}$ with $A^c = \bigcup_{i=1}^m B_i$. Thus $\Omega = A \cup A^c = A \cup B_1 \cup \dots \cup B_m \in \mathcal{A}(\mathcal{C})$.

(b) Let $A \in \mathcal{A}(\mathcal{C})$, say $A = \bigcup_{i=1}^k B_i$ with $B_i \in \mathcal{C}$. For each i , there exist $B_{i1}, \dots, B_{in_i} \in \mathcal{C}$ with $B_i^c = \bigcup_{j=1}^{n_i} B_{ij}$. Then

$$A^c = \bigcap_{i=1}^k B_i^c = \bigcap_{i=1}^k \bigcup_{j=1}^{n_i} B_{ij} = \bigcup_{j_1=1}^{n_1} \dots \bigcup_{j_k=1}^{n_k} \left(\bigcap_{i=1}^k B_{ij_i} \right).$$

The last equality holds because $x \in \bigcap_{i=1}^k \bigcup_{j=1}^{n_i} B_{ij}$ iff for each i there is some j_i with $x \in B_{ij_i}$, i.e., iff $x \in \bigcap_{i=1}^k B_{ij_i}$ for some $(j_1, \dots, j_k)$.

Since $\mathcal{C}$ is closed under pairwise intersections, by induction it is closed under finite intersections, so each $\bigcap_{i=1}^k B_{ij_i} \in \mathcal{C}$. The union ranges over $n_1 n_2 \dots n_k < \infty$ index tuples, so $A^c \in \mathcal{A}(\mathcal{C})$.

(c) Let $A = \bigcup_{i=1}^k B_i$ and $D = \bigcup_{j=1}^m D_j$ be in $\mathcal{A}(\mathcal{C})$. Then $A \cup D = B_1 \cup \dots \cup B_k \cup D_1 \cup \dots \cup D_m$ is a union of $k + m$ sets in $\mathcal{C}$, so $A \cup D \in \mathcal{A}(\mathcal{C})$.

Thus $\mathcal{A}(\mathcal{C})$ is an algebra.

Part 3: WTS $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$.

Let $\mathcal{E}$ be an algebra on Ω with $\mathcal{C} \subset \mathcal{E}$, and let $A = \bigcup_{i=1}^k B_i \in \mathcal{A}(\mathcal{C})$. Then $B_i \in \mathcal{E}$ for all i . Since $\mathcal{E}$ is closed under pairwise unions, by induction it is closed under finite unions, so $A \in \mathcal{E}$. Thus $\mathcal{A}(\mathcal{C}) \subset \mathcal{E}$. //

(ii)

Show also that $\sigma\langle \mathcal{C} \rangle = \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$.

Answer

Proof.

Part 1: WTS $\sigma\langle \mathcal{C} \rangle \subset \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$.

By (i), $\mathcal{C} \subset \mathcal{A}(\mathcal{C}) \subset \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$. So $\sigma\langle \mathcal{A}(\mathcal{C}) \rangle$ is a σ -algebra containing $\mathcal{C}$, and hence $\sigma\langle \mathcal{C} \rangle \subset \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$.

Part 2: WTS $\sigma\langle \mathcal{A}(\mathcal{C}) \rangle \subset \sigma\langle \mathcal{C} \rangle$.

$\sigma\langle \mathcal{C} \rangle$ is a σ -algebra, and hence an algebra, containing $\mathcal{C}$. By (i), $\mathcal{A}(\mathcal{C})$ is the smallest such algebra, so $\mathcal{A}(\mathcal{C}) \subset \sigma\langle \mathcal{C} \rangle$. Hence $\sigma\langle \mathcal{A}(\mathcal{C}) \rangle \subset \sigma\langle \mathcal{C} \rangle$.

Thus $\sigma\langle \mathcal{C} \rangle = \sigma\langle \mathcal{A}(\mathcal{C}) \rangle$. //

Q6: 1.20

Let μ^* be as in (3.1) of Section 1.3:

Given a measure μ on a semialgebra $\mathcal{C} \subset \mathcal{P}(\Omega)$, the outer measure induced by μ is the function μ^* , defined on $\mathcal{P}(\Omega)$ as:

$$\mu^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C}, A \subset \bigcup_{n \geq 1} A_n \right\}$$

(**Hint:** Fix $0 < \epsilon < \infty$. If $\mu^*(A_n) < \infty$ for all $n \in \mathbb{N}$, then find, for each n , a cover $\{A_{nj}\}_{j \geq 1} \subset \mathcal{C}$, $A_n \subset \bigcup_{j \geq 1} A_{nj}$ such that $\sum_{j=1}^{\infty} \mu(A_{nj}) \leq \mu^*(A_n) + \frac{\epsilon}{2^n}$.)

Verify:

(3.4)

$$\mu^*(\emptyset) = 0$$

Answer

Notation. For $A \subset \Omega$, let

$$\Sigma(A) \equiv \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C}, A \subset \bigcup_{n \geq 1} A_n \right\},$$

so that $\mu^*(A) = \inf \Sigma(A)$. Since μ takes values in $[0, \infty]$, every element of $\Sigma(A)$ is nonnegative, and hence $\mu^*(A) \geq 0$ for all $A \subset \Omega$.

Proof.

Step 1: $\emptyset \in \mathcal{C}$. Assume $\mathcal{C} \neq \emptyset$, as in Q5, and pick any $A \in \mathcal{C}$. By the definition of a semialgebra, there exist $B_1, \dots, B_k \in \mathcal{C}$ with $A^c = \bigcup_{i=1}^k B_i$. Since $\mathcal{C}$ is closed under intersections, $A \cap B_1 \in \mathcal{C}$. But $A \cap B_1 \subset A \cap A^c = \emptyset$, so $\emptyset = A \cap B_1 \in \mathcal{C}$.

Step 2. Take $A_n = \emptyset \in \mathcal{C}$ for all $n \geq 1$. Then $\emptyset \subset \bigcup_{n \geq 1} A_n$, and since μ is a measure on $\mathcal{C}$, $\mu(\emptyset) = 0$. Thus

$$0 \leq \mu^*(\emptyset) \leq \sum_{n=1}^{\infty} \mu(\emptyset) = 0,$$

and so $\mu^*(\emptyset) = 0$. //

(3.5)

$$A \subset B \Rightarrow \mu^*(A) \leq \mu^*(B)$$

Answer

Proof. Suppose $A \subset B$, and let $\{A_n\}_{n \geq 1} \subset \mathcal{C}$ with $B \subset \bigcup_{n \geq 1} A_n$. Then $A \subset B \subset \bigcup_{n \geq 1} A_n$, so $\sum_{n=1}^{\infty} \mu(A_n) \in \Sigma(A)$. Thus $\Sigma(B) \subset \Sigma(A)$.

The infimum over a larger set is no greater than the infimum over a smaller one, so

$$\mu^*(A) = \inf \Sigma(A) \leq \inf \Sigma(B) = \mu^*(B). \quad //$$

(3.6)

For any $\{A_n\}_{n \geq 1} \subset \mathcal{P}(\Omega)$,

$$\mu^* \left(\bigcup_{n \geq 1} A_n \right) \leq \sum_{n=1}^{\infty} \mu^*(A_n).$$

Answer

Proof. Let $\{A_n\}_{n \geq 1} \subset \mathcal{P}(\Omega)$ be given, and set $A \equiv \bigcup_{n \geq 1} A_n$.

Case 1: $\mu^*(A_{n_0}) = \infty$ for some n_0 . Then $\sum_{n=1}^{\infty} \mu^*(A_n) = \infty$, and the inequality holds trivially.

Case 2: $\mu^*(A_n) < \infty$ for all $n \geq 1$. Fix $0 < \epsilon < \infty$. For each n , $\mu^*(A_n) + \epsilon/2^n > \mu^*(A_n) = \inf \Sigma(A_n)$, so it is not a lower bound of $\Sigma(A_n)$. Thus there exists $\{A_{nj}\}_{j \geq 1} \subset \mathcal{C}$ with $A_n \subset \bigcup_{j \geq 1} A_{nj}$ and

$$\sum_{j=1}^{\infty} \mu(A_{nj}) \leq \mu^*(A_n) + \frac{\epsilon}{2^n}.$$

Now $\{A_{nj} : n, j \geq 1\} \subset \mathcal{C}$ is countable, since it is indexed by $\mathbb{N} \times \mathbb{N}$, so it can be enumerated as a single sequence in $\mathcal{C}$. It also covers A , since $A = \bigcup_{n \geq 1} A_n \subset \bigcup_{n \geq 1} \bigcup_{j \geq 1} A_{nj}$.

Since the terms $\mu(A_{nj})$ are nonnegative, their sum does not depend on the order of summation. Thus

$$\mu^*(A) \leq \sum_{n=1}^{\infty} \sum_{j=1}^{\infty} \mu(A_{nj}) \leq \sum_{n=1}^{\infty} \left(\mu^*(A_n) + \frac{\epsilon}{2^n} \right) = \sum_{n=1}^{\infty} \mu^*(A_n) + \epsilon.$$

Since $\epsilon > 0$ is arbitrary, $\mu^*(A) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$. //

Q7: 1.24

Verify that $\mathcal{C}_2$, defined in (3.11), is a semialgebra.

Answer

Recall from (3.7) and (3.11):

$$\mathcal{C} \equiv \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\}, \quad \mathcal{C}_2 \equiv \{I_1 \times I_2 : I_1, I_2 \in \mathcal{C}\}.$$

Proof.

Part 0: $\mathcal{C}$ is a semialgebra.

Intersections. For $-\infty \leq a, c$ and $b, d < \infty$:

$$(a, b] \cap (c, d] = (a \vee c, b \wedge d], \quad (a, b] \cap (c, \infty) = (a \vee c, b], \quad (a, \infty) \cap (c, \infty) = (a \vee c, \infty),$$

where an interval with left endpoint greater than or equal to its right endpoint is $\emptyset = (0, 0] \in \mathcal{C}$. All of these are in $\mathcal{C}$.

Complements. $(a, b]^c = (-\infty, a] \cup (b, \infty)$ and $(a, \infty)^c = (-\infty, a]$. These are finite disjoint unions of sets in $\mathcal{C}$, where $(-\infty, a] \in \mathcal{C}$ because (3.7) allows a left endpoint of $-\infty$. (If $a = -\infty$, then $(-\infty, a] = \emptyset \in \mathcal{C}$.)

Part 1: WTS $A, B \in \mathcal{C}_2 \Rightarrow A \cap B \in \mathcal{C}_2$.

Let $A = I_1 \times I_2$ and $B = I_3 \times I_4$ with $I_1, \dots, I_4 \in \mathcal{C}$. Then

$$A \cap B = (I_1 \cap I_3) \times (I_2 \cap I_4),$$

since $(x, y) \in A \cap B$ iff $x \in I_1 \cap I_3$ and $y \in I_2 \cap I_4$. By Part 0, $I_1 \cap I_3, I_2 \cap I_4 \in \mathcal{C}$, so $A \cap B \in \mathcal{C}_2$.

Part 2: WTS for $A \in \mathcal{C}_2$, A^c is a finite disjoint union of sets in $\mathcal{C}_2$.

Let $A = I_1 \times I_2 \in \mathcal{C}_2$. A point $(x, y) \in A^c$ iff $x \notin I_1$ or $y \notin I_2$. Splitting on which of these hold,

$$A^c = (I_1^c \times I_2) \cup (I_1 \times I_2^c) \cup (I_1^c \times I_2^c),$$

and the three pieces are pairwise disjoint, since they differ in whether $x \in I_1$ or $y \in I_2$.

By Part 0, for $k = 1, 2$, there exist disjoint $I_{k1}, \dots, I_{kn_k} \in \mathcal{C}$ with $I_k^c = \bigcup_{j=1}^{n_k} I_{kj}$. Since products distribute over unions,

$$I_1^c \times I_2 = \bigcup_{j=1}^{n_1} (I_{1j} \times I_2), \quad I_1 \times I_2^c = \bigcup_{l=1}^{n_2} (I_1 \times I_{2l}), \quad I_1^c \times I_2^c = \bigcup_{j=1}^{n_1} \bigcup_{l=1}^{n_2} (I_{1j} \times I_{2l}).$$

Each set on the right is a product of sets in $\mathcal{C}$, and hence is in $\mathcal{C}_2$.

- *Disjoint within each piece:* two products are disjoint whenever their first coordinates (or their second coordinates) are disjoint, and the I_{1j} are pairwise disjoint, as are the I_{2l} .
- *Disjoint across pieces:* the three pieces are pairwise disjoint.

Thus A^c is a disjoint union of $n_1 + n_2 + n_1 n_2 < \infty$ sets in $\mathcal{C}_2$.

By Parts 1 and 2, $\mathcal{C}_2$ is a semialgebra. //

Q8: Extra

Consider the following function:

$$F(x) = x \mathbb{I}\{0 < x < 1\} + \mathbb{I}\{x \geq 1\}, \quad x \in \mathbb{R}$$

(i)

Is this $F(\cdot)$ non-decreasing, right-continuous? Check.

Answer

Yes. In fact, F is non-decreasing and continuous, and hence right-continuous.

Proof. We claim $F(x) = \min\{\max\{x, 0\}, 1\}$ for all $x \in \mathbb{R}$.

Case 1: $x \leq 0$. Then $\max\{x, 0\} = 0$, so the right side is $0 = F(x)$.

Case 2: $0 < x < 1$. Then $\max\{x, 0\} = x < 1$, so the right side is $x = F(x)$.

Case 3: $x \geq 1$. Then $\max\{x, 0\} = x \geq 1$, so the right side is $1 = F(x)$.

Let $g(x) \equiv \max\{x, 0\} = \frac{x+|x|}{2}$ and $h(y) \equiv \min\{y, 1\} = \frac{y+1-|y-1|}{2}$. Both are continuous, since $|\cdot|$ is continuous. Both are also non-decreasing, since for $x_1 \leq x_2$ we have $\max\{x_1, 0\} \leq \max\{x_2, 0\}$ and $\min\{x_1, 1\} \leq \min\{x_2, 1\}$.

Thus $F = h \circ g$ is non-decreasing (a composition of non-decreasing functions) and continuous (a composition of continuous functions). In particular, $F(x+) = F(x)$ for all $x \in \mathbb{R}$, so F is right-continuous. //

(ii)

Define suitably a Lebesgue-Stieltjes measure μ_F on a suitable semialgebra $\mathcal{C}$. Do not have to prove $\mathcal{C}$ is a semi-algebra or μ_F is a measure.

Answer

Take $\mathcal{C}$ to be the semialgebra in (3.7):

$$\mathcal{C} \equiv \{(a, b] : -\infty \leq a \leq b < \infty\} \cup \{(a, \infty) : -\infty \leq a < \infty\}.$$

Set $F(-\infty) \equiv \lim_{x \rightarrow -\infty} F(x) = 0$ and $F(\infty) \equiv \lim_{x \rightarrow \infty} F(x) = 1$. Following (3.8), define

$$\mu_F((a, b]) = F(b+) - F(a+) = F(b) - F(a), \quad \mu_F((a, \infty)) = F(\infty) - F(a+) = 1 - F(a),$$

where $F(x+) = F(x)$ by (i).

Equivalently, since $F(x)$ is the length of $(0, x] \cap (0, 1)$, the measure of $I \in \mathcal{C}$ is the length of $I \cap (0, 1)$.

(iii)

Prove that $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$. (Can use the fact that $\mathcal{B}(\mathbb{R}) = \sigma\langle\{\text{open sets}\}\rangle = \sigma\langle\mathcal{O}_1\rangle = \sigma\langle\mathcal{O}_2\rangle = \sigma\langle\mathcal{O}_3\rangle = \sigma\langle\mathcal{O}_4\rangle$ as in 1.10.)

Answer

Here, for $k = 1$, $\mathcal{O}_1 = \{(a, b) : -\infty \leq a < b \leq \infty\}$ and $\mathcal{O}_2 = \{(-\infty, x) : x \in \mathbb{R}\}$.

Proof.

Part 1: WTS $\mathcal{B}(\mathbb{R}) \subset \sigma\langle\mathcal{C}\rangle$.

Let $x \in \mathbb{R}$. We claim $(-\infty, x) = \bigcup_{n \geq 1} (-\infty, x - n^{-1}]$.

($\supset$) Each $(-\infty, x - n^{-1}] \subset (-\infty, x)$.

($\subset$) If $y < x$, choose n with $n^{-1} \leq x - y$. Then $y \leq x - n^{-1}$.

Each $(-\infty, x - n^{-1}] \in \mathcal{C}$ (take $a = -\infty$), so $(-\infty, x) \in \sigma\langle\mathcal{C}\rangle$. Thus $\mathcal{O}_2 \subset \sigma\langle\mathcal{C}\rangle$, and so $\mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{O}_2\rangle \subset \sigma\langle\mathcal{C}\rangle$.

Part 2: WTS $\sigma\langle\mathcal{C}\rangle \subset \mathcal{B}(\mathbb{R})$.

Sets (a, ∞) : $(a, \infty) \in \mathcal{O}_1$ directly (take $b = \infty$).

Sets $(a, b]$ with $a = b$: $(a, a] = \emptyset \in \mathcal{B}(\mathbb{R})$.

Sets $(a, b]$ with $-\infty \leq a < b < \infty$: We claim $(a, b] = \bigcap_{n \geq 1} (a, b + n^{-1})$.

($\subset$) $b < b + n^{-1}$ for all n .

($\supset$) If $a < y < b + n^{-1}$ for all n , then $y \leq b$.

Each $(a, b + n^{-1}) \in \mathcal{O}_1$, and $\sigma\langle\mathcal{O}_1\rangle$ is closed under countable intersections, so $(a, b] \in \sigma\langle\mathcal{O}_1\rangle$.

Thus $\mathcal{C} \subset \sigma\langle\mathcal{O}_1\rangle = \mathcal{B}(\mathbb{R})$, and so $\sigma\langle\mathcal{C}\rangle \subset \mathcal{B}(\mathbb{R})$.

Therefore $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$. //

(iv)

Argue that there exist a μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ that extends μ_F defined above. Do not need to prove this step - just state a result for your answer.

Answer

By Caratheodory's extension theorem (Theorem 1.3.3), since μ_F is a measure on the semialgebra $\mathcal{C}$, the outer measure μ_F^* induced by μ_F via (3.1) satisfies the following:

- $\mathcal{M}_{\mu_F^*}$ is a σ -algebra and μ_F^* is a measure on it (Theorem 1.3.2);

- $\mathcal{C} \subset \mathcal{M}_{\mu_F^*}$;
- $\mu_F^* = \mu_F$ on $\mathcal{C}$.

Since $\mathcal{M}_{\mu_F^*}$ is a σ -algebra containing $\mathcal{C}$, (iii) gives $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C}) \subset \mathcal{M}_{\mu_F^*}$. Thus $\mu \equiv \mu_F^*|_{\mathcal{B}(\mathbb{R})}$ is a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with $\mu = \mu_F$ on $\mathcal{C}$.

(v)

Is this extension unique? (i.e. if μ' is another extension of μ_F , then is it true that $\mu(A) = \mu'(A)$, for all $A \in \mathcal{B}(\mathbb{R})$?). Justify your answer (no proof needed). What is the “common name” for this measure μ ?

Answer

Yes, the extension is unique.

Note that $\mathbb{R} = (-\infty, \infty) \in \mathcal{C}$ and $\mu_F(\mathbb{R}) = F(\infty) - F(-\infty) = 1$. So any extension μ' of μ_F satisfies $\mu'(\mathbb{R}) = \mu(\mathbb{R}) = 1$, and both μ and μ' are finite. Moreover, $\mathcal{C}$ is a π -system, since a semialgebra is closed under intersections by definition, and $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R})$ by (iii).

Thus, by the uniqueness of measures (Theorem 1.2.4), $\mu = \mu'$ on $\mathcal{B}(\mathbb{R})$. Equivalently, since μ_F is finite and hence σ -finite on $\mathcal{C}$, Theorem 1.3.6 gives the same conclusion.

The measure μ is the **Uniform(0, 1) distribution**: the probability measure with CDF F . Equivalently, it is Lebesgue measure restricted to $(0, 1)$, i.e., $\mu(A) = m(A \cap (0, 1))$ for $A \in \mathcal{B}(\mathbb{R})$.